

NATIONAL CYBERSECURITY

India has a need to secure the assets of the country in the virtual realm as well. Currently, this is being done by the National Technical Research Organisation, the Defense Intelligence Agency, the National Information Infrastructure Protection Centre, and the Indian Computer Emergency Response Team. India does not have a cohesive approach, a cyber army if you will, like the one that China has. The importance of such a cohesive approach is only going to escalate with the advent of IoT devices, and the progress of the smart cities and Digital India initiatives. There are plans to rectify the situation, though. The government intends to start a new tri-service agency dedicated to this task. The agency will be staffed with over 1,000 personnel from all the three arms of the forces – army, navy and air force. The idea is to secure the critical infrastructure of India from cyber attacks. The Defence Cyber Agency will have both offensive and defensive capabilities. The agency will work closely with the National Cyber Security Advisor, and tackle all non-civilian problems that the country faces, in terms of cybersecurity. This includes cyber attacks, cyber terrorism, cyber warfare, and cyber espionage. The agency will also be needed to tackle cyber propaganda as well, such as the campaign unleashed by China during the Doklam standoff.

systems, along with the tether itself have also been indigenously developed. The winch mooring system and the gas management system are both mobile, mounted on the backs of trucks. The aerostat is outfitted with state of the art imaging technologies, including infrared cameras for night vision, and HD video cameras for surveillance. The aerostat can remain in the air for a period of 14 days before it needs a refill of the helium that keeps it afloat. The balloon was realised in collaboration with private industries, and its development is a part of the Make in India initiative. The aerostat can be configured to work as an airborne early warning system, and is capable of detecting low flying aircraft or missiles. The team that developed the Nakshatra is now working on Nakshatra-II, an enhanced version of the aerostat.

RUSTOM-II

The Rustom-II is a Medium Altitude Long Endurance (MALE) UAV. The UAV is being developed for Intelligence, Surveillance and Reconnaissance (ISR) capabilities. The drone is capable of carrying various payloads for these purposes. DRDO labs, in collaboration with private industries are indigenously developing these payloads. The indigenous development of these payloads is a part of the Make in India initiative. The airframe,

landing gear, flight control subsystem, and avionics subsystem are all entirely produced in India. A new Aeronautical Test Range (ATR) has been set up in Chitradurga in Karnataka to test out the capabilities of Indian UAVs. The capabilities of the flying platform such as take off, level flight, banking and landing have all been proven in a test flight at this facility in November 2016. The Rustom-I can fly at an altitude of 15,000 feet above sea level, and has an endurance of 12 hours. The Rustom-II is designed to be twice as good, with the capability of flying at 30,000 feet above sea level, and an endurance of 24 hours from take off. 7 airframes have been realised so far. In the future, the UAV can also be made combat capable, with weaponised payloads fitted on board.

TEJAS

In the 1950s, engineers at Hindustan Aircraft Limited (HAL) were interested in developing a supersonic fighter aircraft. The Indian Government authorised the development of the aircraft, and the result was the HAL HF-24 Marut. Due to various reasons, the capacity of the engines were limited, and the aircraft could never achieve supersonic speeds. Still, it was the first indigenously developed fighter air-

craft to go into production and active service. The HF-24 Marut entered into active service as a fighter-bomber. A total of 147 aircraft were produced, and the assembly line was shut down.

By the early 80s, there was no production facility in the country that was capable of developing a light combat aircraft (LCA) on its own. The DRDO had to develop an aircraft from scratch, which is why the LCA program was started. The intention was not just to develop and produce the aircraft locally, but to build up the local industries, and eventually allow them to produce commercial products for the global market. The first technology demonstration of this program took flight in 2001. The then Prime Minister, Atal Bihari Vajpayee, named the aircraft “Tejas” - which means radiance in Sanskrit.

Since that time, over 3,000 flights in a total of 16 Tejas aircraft have been completed. The lightweight supersonic aircraft is capable of filling multiple roles in combat. It can carry four tonnes of weapons, fire air to air missiles, or drop laser guided bombs. Tejas is a single seater, single engine aircraft. There are thirteen different Tejas variants undergoing flight trials, including a version for the Indian navy. In January 2016, the aircraft made its international debut



The HAL Tejas flying with the national colours

and showed off some impressive manoeuvres at the Bahrain International Air Show, flying alongside the F-16 Fighting Falcon and the Eurofighter Typhoon. Six months later, the aircraft was formally inducted to the Indian Air Force, with two aircraft forming the Flying Daggers 45 squadron. There are efforts underway to boost up the production capabilities of HAL, to be able to roll out 8 aircrafts every year. 